Social Science Simulations (SVUH.00.300) · Seminar

From Words to Numbers: pseudo-quantification and the quantification of uncertainty

ACHARYA Kishor · Institute of Social Studies, University of Tartu · 07/11/2026

Three things a simulation might need, and nobody can hand you

A perception

How corrupt do people think their institutions are? Surveys give a 1-to-10 answer. The model needs a probability that an agent pays a bribe.

A norm

Why do most people pay the bus fare when checks are rare? Because others do, and because they think others expect it. The model needs a threshold per person.

A number nobody measured

How often is a ticket actually checked? No survey, no statistic. The model needs a value anyway.

Where the last seminar left us, and the two words in the title

The Monte Carlo recipe never changed: a model with something uncertain in it, a random draw for each uncertain thing, repeat and count. But every distribution came from nowhere in particular: we said polls miss by three points; we said a household wants 30% similar neighbours. The closing question was: where did 30% come from?

Pseudo-quantification

Turning qualitative material into numbers a model can use. “Pseudo” because the number is not measured the way a length is measured: it is assigned, by a rule we choose, and a different rule would give a different number.

Quantification of uncertainty

Writing down how sure we are of that assignment, as a distribution, so that the Monte Carlo loop can carry the doubt through the model instead of hiding it.

Why it matters beyond honesty: once the inputs are distributions across people rather than single numbers, the model starts doing things nobody put in: tipping points, cascades, segregation from mild preferences. Heterogeneity is where emergence comes from, and averaging it away is what kills it. Stop 5 shows this with the two models you already know.

Today: six stops, one case, and two tabs to keep open

1. **The gap**
   what a model needs, what research gives
2. **From scales**
   an ordinal answer becomes a threshold: four rules
3. **From words and experts**
   coding text; eliciting a number nobody measured
4. **Quantifying the uncertainty**
   sampling, translation, heterogeneity, structural
5. **Into the model**
   one changed line; self-correction and a tipping point
6. **Reporting**
   what to report, and your project

Tab 1: the tour

acharyakishor.com/simulations
“From words to numbers”.
Sliders, no code.

Tab 2: notebook

github.com/ACHARYAkishor/from_words_to_number_Seminar_UniTartu
(Open in Colab button)

One case all day: fare dodging on Tartu buses as a social norm, with real World Values Survey answers from Estonia and your own questionnaire. Break after Stop 3.

One case for the whole seminar

Fare dodging as a social norm

Why do most people pay when the chance of being checked is small?

Three sources of data today

World Values Survey

Estonia, 1996 (n = 1,021) and 2011 (n = 1,533). “Is avoiding a fare on public transport justifiable?”, 1 never … 10 always. Also taxes and bribes. Published frequency tables.

The class questionnaire

Twelve questions, 21 of you, anonymous. Four about the fare norm, four direct thresholds in four domains, three scales, one sentence of text, one count, one category.

Two models you know

Schelling's city and Granovetter's crowd, unchanged from last seminar except for one line: the single threshold becomes one draw per agent.

Inglehart, R. et al. (eds.) (2014). World Values Survey: Round Three and Round Six, Country-Pooled Datafiles. Madrid: JD Systems Institute. Frequency tables from catalog.ihsn.org (9087, 9049).

What makes a norm a norm (Bicchieri, 2017)

- A social norm is a behaviour people prefer to follow on condition that enough others follow it (empirical expectation) and enough others think they should (normative expectation).
- A habit or a moral conviction does not depend on others; a norm does. That conditionality is what a model can use: each person has a threshold, the share of others at which they would switch.
- The class questionnaire you filled in measured each part, for fare dodging, with ten passengers as the unit:

Question	What it measures	In the theory
Q1 · of 10 passengers, how many dodge?	a belief about others' behaviour	empirical expectation
Q2 · of 10 residents, how many would disapprove?	a belief about others' approval	normative expectation
Q3 · is dodging justifiable, 1 (never) … 10 (always)?	own attitude, on a scale	personal belief (the WVS item)
Q4 · at how many dodgers of 10 would you dodge?	own switching point, in the model's units	conditional preference = the threshold

Bicchieri, C. (2017). Norms in the Wild: How to Diagnose, Measure, and Change Social Norms. Oxford University Press.

Stop 1

The gap: what a model needs, what research gives

Four kinds of number on one side, five kinds of material on the other

Four kinds of number a simulation needs

Kind	In Schelling's city	In Granovetter's crowd	In the referendum
Agent attributes	each household's threshold and kind	each person's threshold	—
Rules	move if below threshold, to a random empty house	join once the share of others reaches my threshold	passes if yes > 50%
Parameters	grid size, share of empty houses	crowd size	poll SD 3, turnout mean −1
Initial state	the random starting arrangement	nobody has joined	polled share 52%

Rules are written in words first and then as "if" statements; attributes and parameters are numbers; the initial state is a mix. Pseudo-quantification is mostly about attributes and parameters, but a rule can hide a number too: "most of my neighbours" is a rule with a threshold in it.

Five kinds of material research gives (Stevens, 1946)

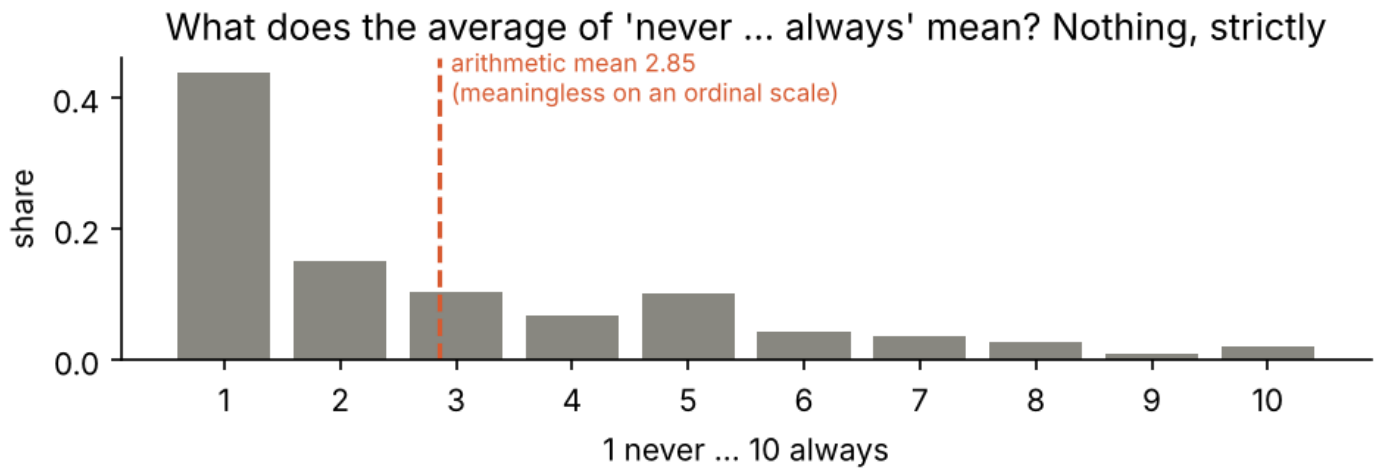
Level	What it means	Operations that make sense	Examples	Translation needed
Nominal	categories with no order; a label	count; mode; same or different	occupation; bus user / non-user	code to indicators; never add them
Ordinal	a meaningful order, but unknown or unequal distances	rank; median; more or less than	race positions; Likert items; “never … always” 1–10	the whole of Stop 2
Interval	equal distances, no true zero	add; subtract; mean	temperature in °C; a calendar year	little
Ratio	equal distances and a true zero	everything: ratios too	cups of coffee; neighbours out of eight	none
Not a number	a sentence, a judgement, a relation	none until coded	an interview answer; an expert's guess	Stop 3

Stevens, S. S. (1946). On the theory of scales of measurement. Science, 103(2684), 677–680.

Your twelve questions, sorted by what each one is

Q	The question (short)	Level of measurement	What it is in the theory	Simulation input
1	How many of 10 passengers dodge?	count 0–10 (a belief)	empirical expectation	attribute
2	How many of 10 residents would disapprove?	count 0–10 (a belief)	normative expectation	attribute
3	Is dodging justifiable? 1–10	ordinal scale	personal normative belief (the WVS item)	attribute: needs a rule
4	At how many of 10 dodgers would you dodge? (+ never)	threshold 0–10	conditional preference	attribute: direct
5	How many of 8 neighbours should be like you?	threshold 0–8	Schelling threshold	attribute: direct
6	Uncomfortable if most neighbours differ, 1–5	ordinal scale	same attitude as Q5	attribute: needs a rule
7	Cups of coffee per day	ratio	control: a true number	—
8	Loud movie nights before complaining (+ never)	threshold 0–10	a threshold, trivial domain	attribute: direct
9	Friends of 10 before adopting an app	threshold 0–10	Granovetter threshold	attribute: direct
10	Trust in the bus timetable, 1–5	ordinal scale	attitude with no behavioural anchor	attribute: hard to map
11	In one sentence: why?	text	qualitative material	Stop 3
12	How often do you use a bus or Smart Bike?	ordered category	a grouping variable	rule / sub-population

The mean of an ordinal scale means nothing; the shape means a lot



- 2011, Estonia, “avoiding a fare is justifiable”, 1,503 answers.
- The arithmetic mean, 2.85, is drawn in. Nobody has established that the step from 1 to 2 equals the step from 9 to 10, so it has no strict meaning.
- The shape does: 44% said “never”. That one bar will decide more in the model than anything the mean could say.

The chain: concept → indicator → variable → parameter

Link	What it is	The researcher's choice	For fare dodging
Concept	an abstract idea; cannot be observed directly	which theory of the concept to adopt	“fare dodging is a social norm” (Bicchieri)
Indicator	something observable that stands for the concept	which observable things count as evidence	what people believe others do and approve of; what they would do themselves
Variable	the indicator measured, with a scale and units	which question, which scale, which units	Q1–Q4: counts out of 10, a 1–10 scale, a tick-box
Parameter / attribute	the variable turned into what the model consumes	which rule turns the scale into the number (Stop 2)	each agent's threshold, between 0 and 1

Each link narrows the previous one, and each narrowing can be done differently. The chain is written out so a reader who disagrees with the result can see which link to argue with.

Lazarsfeld, P. F. (1958). Evidence and inference in social research. Daedalus, 87(4), 99–130.

The translation recipe: four steps, and one question between them

1 · The material

what research gave you: a category, an ordered scale, a count, a sentence, a judgement

2 · The rule

the stated procedure that turns it into a number: a mapping, a codebook, an elicitation protocol. Always a choice, always written down

3 · The distribution

what the rule produces, across people and across our doubt about the rule. Never a single number

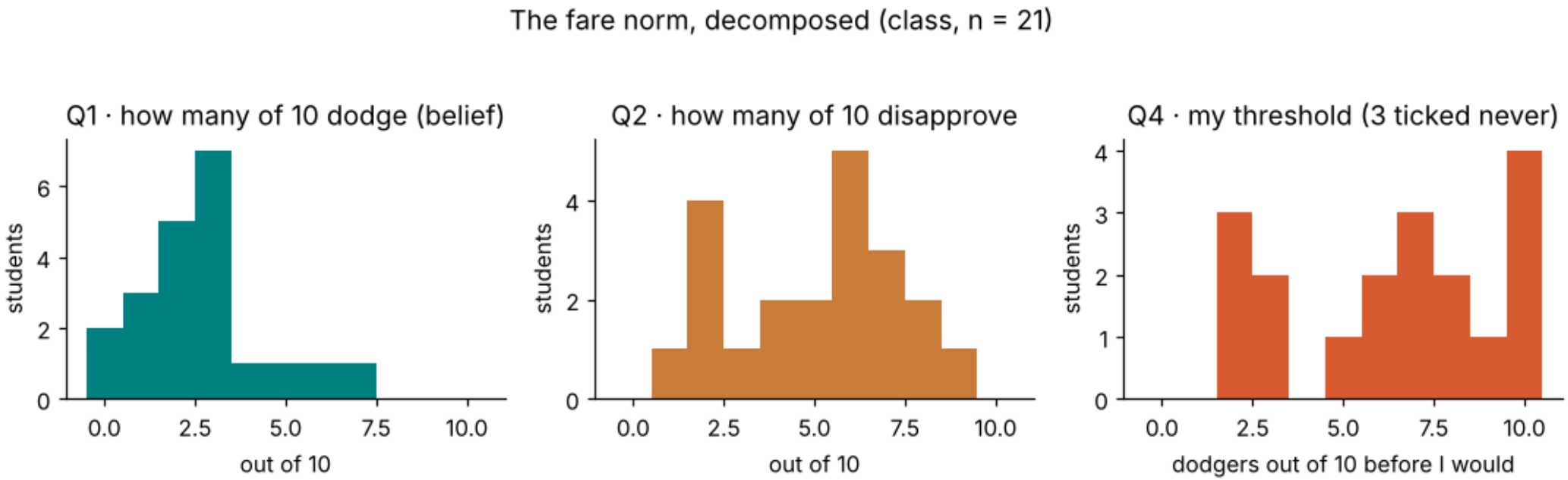
4 · The use

which of the four inputs it becomes: an attribute, a rule, a parameter, an initial state

Between 3 and 4: what uncertainty is carried forward into the model? Each stop ends with a box that names the four steps and the uncertainty for that stop.

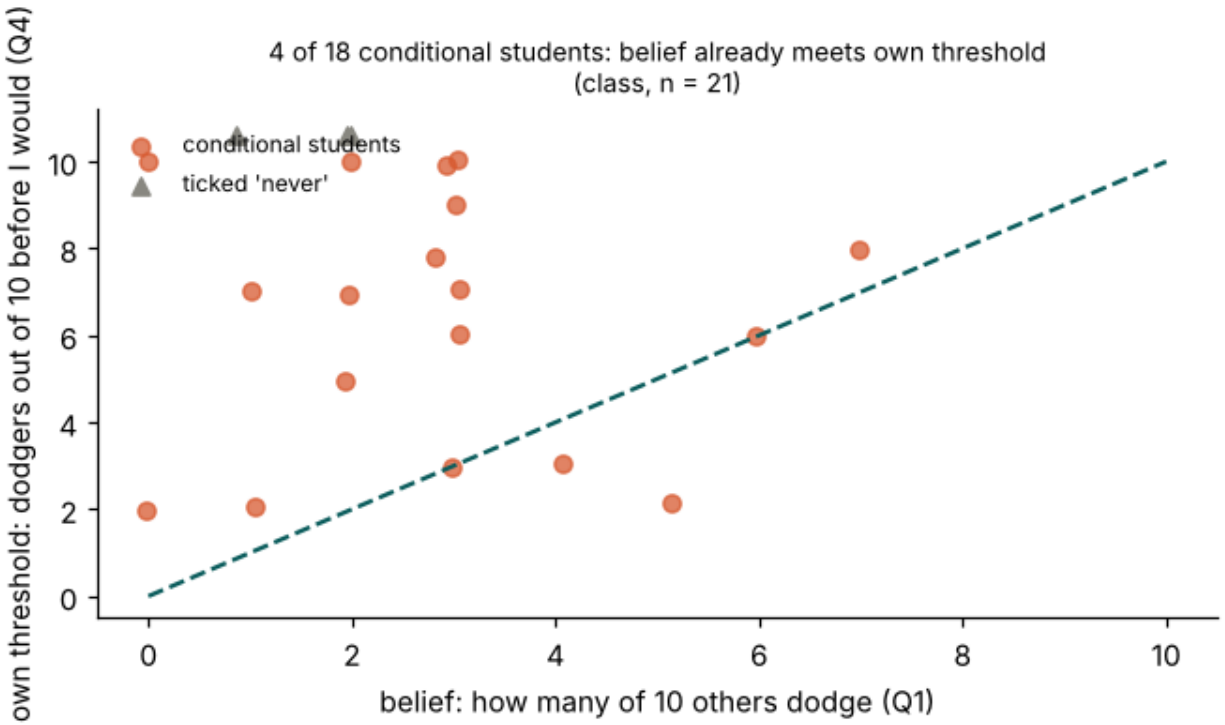
This is my way of keeping the choices in view, not a standard everyone uses. Textbooks call the whole path operationalisation; the four steps are a bookkeeping device for this seminar.

Your answers: the fare norm decomposed



Left: how many of 10 you believe dodge (Q1). Middle: how many of 10 would disapprove (Q2). Right: your own threshold (Q4), excluding those who ticked “never”. Those are the people for whom this is not conditional at all: a conviction, not a norm. Their answer is censored: beyond 10, we do not know where.

A side-glance: is the norm holding, or under strain?



- One dot per student: belief about others across, own threshold up. On the diagonal, belief equals threshold.
- On or below the line: by your own rule you should already be dodging. The norm is not what holds you back.
- A norm is under strain when many people are in that position; it survives only because beliefs happen to sit below thresholds.
- Not a step on the ladder: a static snapshot, each person judged alone. Stop 5 puts the same thresholds in a crowd that reacts to itself.

On your screen: Stop 1

Website tour Stop 1 of the tour: the WVS item, with item and year dropdowns

Notebook Notebook 1.1–1.3: the catalogue of the twelve questions; your answers; the strain plot with its slider

Parameter	What it means	What to watch
Item / Year	Which WVS question, which year	The bribe item is almost all “never”; the fare item has a tail and a bump at 5. Taxes got stricter 1996→2011, fares looser
Believe X/10 dodge (notebook)	Everyone now believes X of 10 others dodge, as after a convincing news story	The dots move right; every dot on or below the diagonal is a student whose condition is now met

Task: find the smallest X at which half the conditional class is released: that is how big a belief shock the norm could survive.

What to take from Stop 1

- A simulation needs four kinds of number; research gives five kinds of material.
- The chain concept → indicator → variable → parameter is where the translation happens, and every link is a choice.
- An ordinal scale is a rank with a label, not a quantity. Its mean is not a measurement; its shape is.

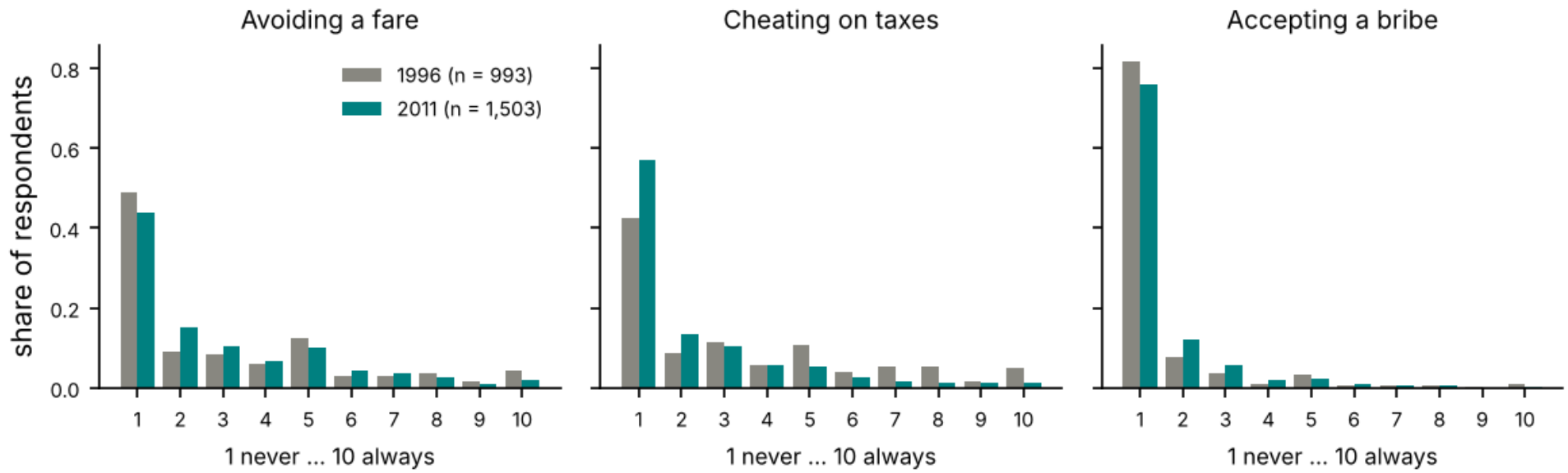
What this stop adds to the argument: *naming the gap. Everything that follows is a way of crossing it, and the recipe box at every stop records how.*

Stop 2

From scales: an ordinal answer becomes a threshold

Four defensible rules, and two ways to avoid needing one

The data: “is it justifiable?”, Estonia 1996 and 2011



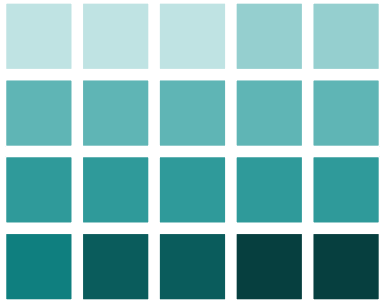
The shapes differ: the bribe item is almost all “never”; the fare item has a long tail and a bump at 5. Between 1996 and 2011 Estonians became stricter about taxes (never: 42% → 57%) and looser about fares (49% → 44%), while the bribe item barely moved. “Justifiable, 1 to 10” is three different kinds of number here.

Inglehart et al. (2014), World Values Survey, Estonia waves 3 and 6. Variables V193/V194/V196 (1996), V199/V201/V202 (2011). Unweighted counts of valid cases.

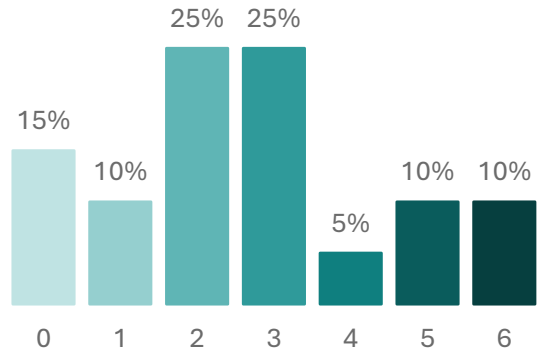
The synthetic population

A survey gives a list of answers; a model needs as many values as it has agents. The bridge: each answer's share becomes its probability, and every agent draws one answer at random, with replacement. It adds no information; it lets the list stand in for the population, as if the list were the population.

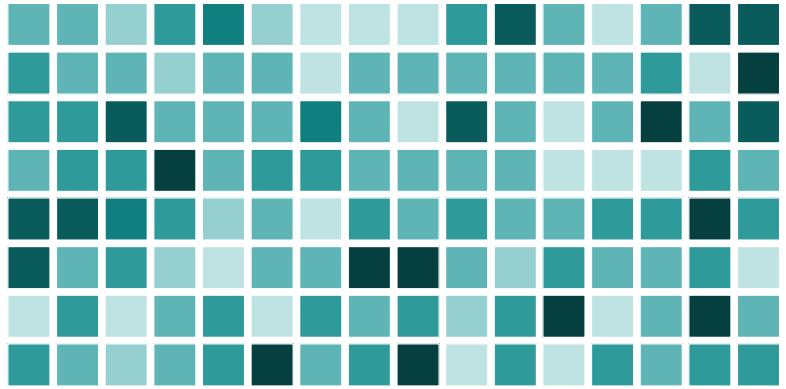
1 · The list: 20 students, Q5 (of 8 like you?)



2 · Count → share → probability



3 · 900 households, each draws one answer



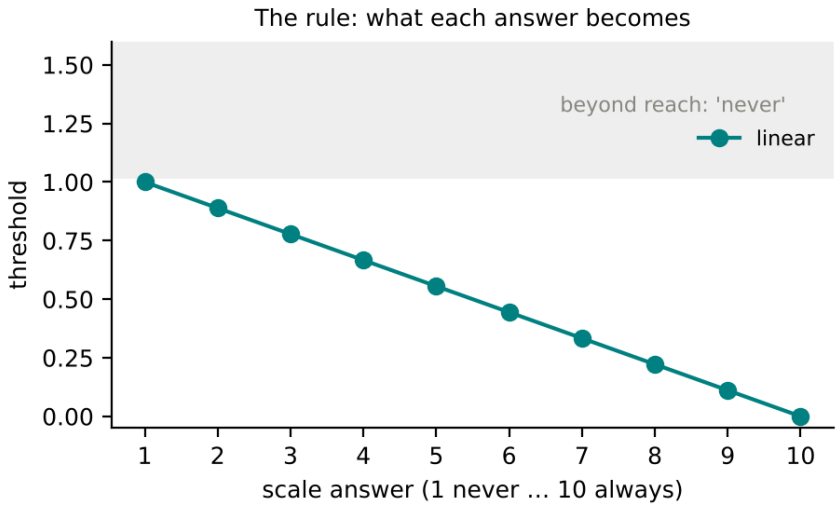
“three of eight” was said by 5 of 20 students, so a household drawn from the class wants three similar neighbours with probability 5/20 = 25%. Each square on the right is a household; its shade is the answer it drew. The shades appear in the same proportions as on the left, and every household's threshold is something a real student said.

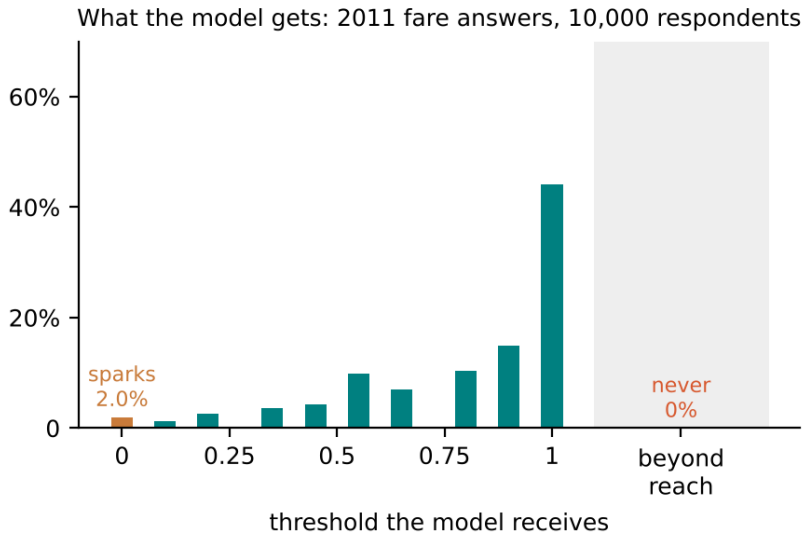
Where we do this: the rule explorer next (10,000 imaginary Estonians from the 1,503 real answers); Stop 4 (surveys of size n drawn the same way: the bootstrap); Stop 5 (the city and the crowd). The doubt that comes from the “as if” is the first of Stop 4's four uncertainties.

Do it yourself now: website tour, Stop 2, “First, the synthetic population”. Press Fill from the class and watch the bars converge; then Fill from the average (clones); then Run the city after each, and compare the two segregation numbers.

Rule 1 · Linear: every step is the same size

answer	10	9	8	7	6	5	4	3	2	1
threshold t = (10 − s)/9	0	0.11	0.22	0.33	0.44	0.56	0.67	0.78	0.89	1.00





0 and 1. t = 0: dodges regardless (the sparks: everyone who answered 10). t = 1: dodges only when 100% of others do. Ten answers, nine steps, each step 1/9 ≈ 0.11 of threshold.

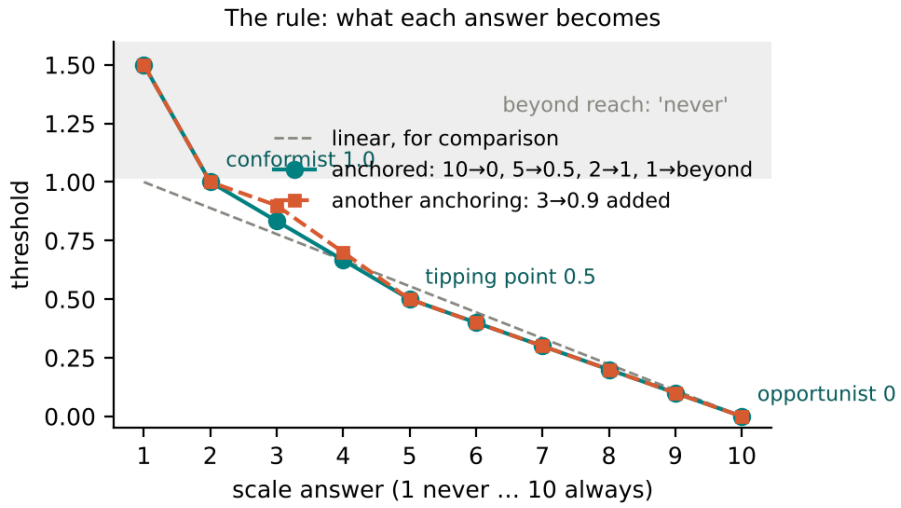
The choice. Equal spacing, which the scale never promised. And "never" (1) lands at 1.0, the same place as "only if everyone does": the two are lumped together and both stay reachable.

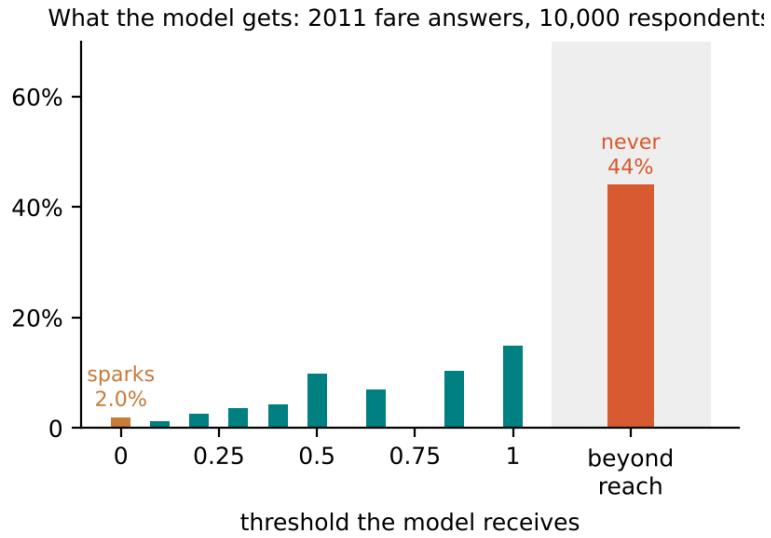
Effect. Nothing hidden, easy to explain. Nobody is beyond reach (0% never), so a cascade that reached 100% would recruit everyone. Sparks 2%.

On a 1–5 scale: four steps of 0.25: 5 → 0, 4 → 0.25, 3 → 0.5, 2 → 0.75, 1 → 1.0. The rule is applied per person: each respondent's answer becomes that respondent's threshold; the counts only say how many people land at each value.

Rule 2 · Anchored: a few answers mean what they say

answer	10	9	8	7	6	5	4	3	2	1
anchored (ours)	0	0.10	0.20	0.30	0.40	0.50	0.67	0.83	1.00	beyond
linear, to compare	0	0.11	0.22	0.33	0.44	0.56	0.67	0.78	0.89	1.00
another anchoring (3 → 0.9)	0	0.10	0.20	0.30	0.40	0.50	0.70	0.90	1.00	beyond





Anchors are judgements. 10 → 0 (always: the opportunist); 5 → 0.5 (torn: follows the majority); 2 → 1.0 (the conformist); 1 → beyond reach (never). The 0.5 was not computed: it was decided.

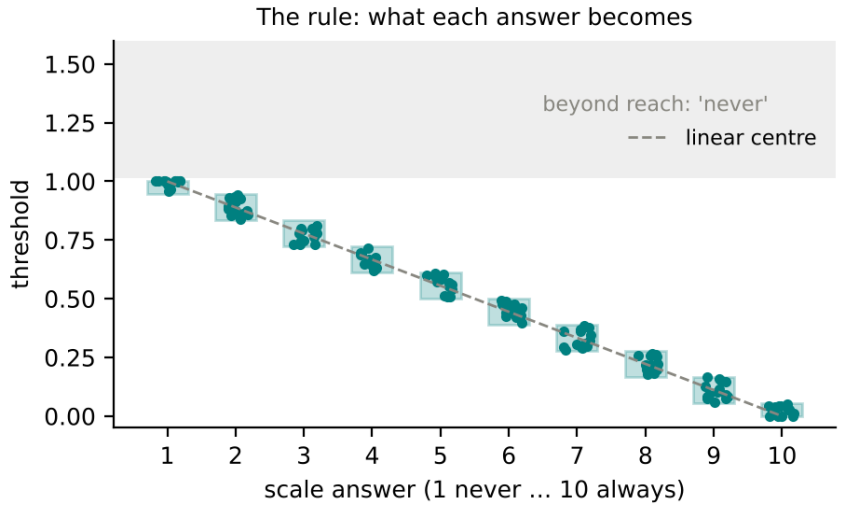
Between anchors: straight lines. 10 to 5: five steps for 0.5 of threshold, 0.10 each. 5 to 2: three steps for 0.5, 0.167 each. Linear is the special case where the only anchors are 10 → 0 and 1 → 1.

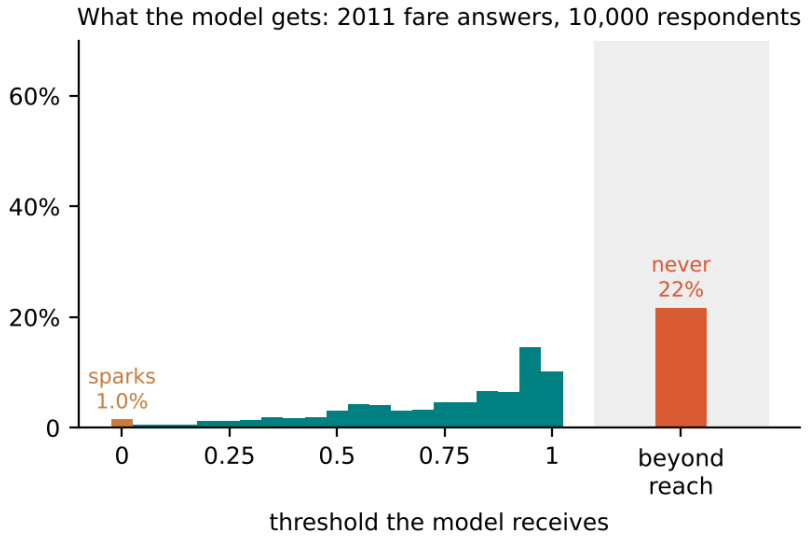
Effect. The upper half of the scale is squeezed, the lower half stretched, and the 1s leave the game: 44% beyond reach. Sparks 2%. Add an anchor at 3 → 0.9 and only 3 and 4 move.

On a 1–5 scale with "1 means never": 5 → 0, 3 → 0.5, 2 → 1.0, 1 → beyond, and 4 → 0.25 by interpolation; it differs from linear only at the two lowest answers. The shorter the scale, the less room for rules to disagree, except at the ends.

Rule 3 · Jitter: an answer is a band, not a point

answer	10	9	8	7	6	5	4	3	2	1
centre (10 – s)/9	0	0.11	0.22	0.33	0.44	0.56	0.67	0.78	0.89	1.00
band: centre ± 1/18	0–0.06	0.06–0.17	0.17–0.28	0.28–0.39	0.39–0.50	0.50–0.61	0.61–0.72	0.72–0.83	0.83–0.94	0.94–beyond





The random part. Start from the linear value; add a uniform random offset between −1/18 and +1/18, half a step either side. So each answer owns a band from halfway to the previous answer to halfway to the next; the bands tile the line.

Per person, no counting. 56 people answered 7 in 2011: 56 random points between 0.28 and 0.39, instead of 56 identical values at 0.33. A person is somewhere in the bucket.

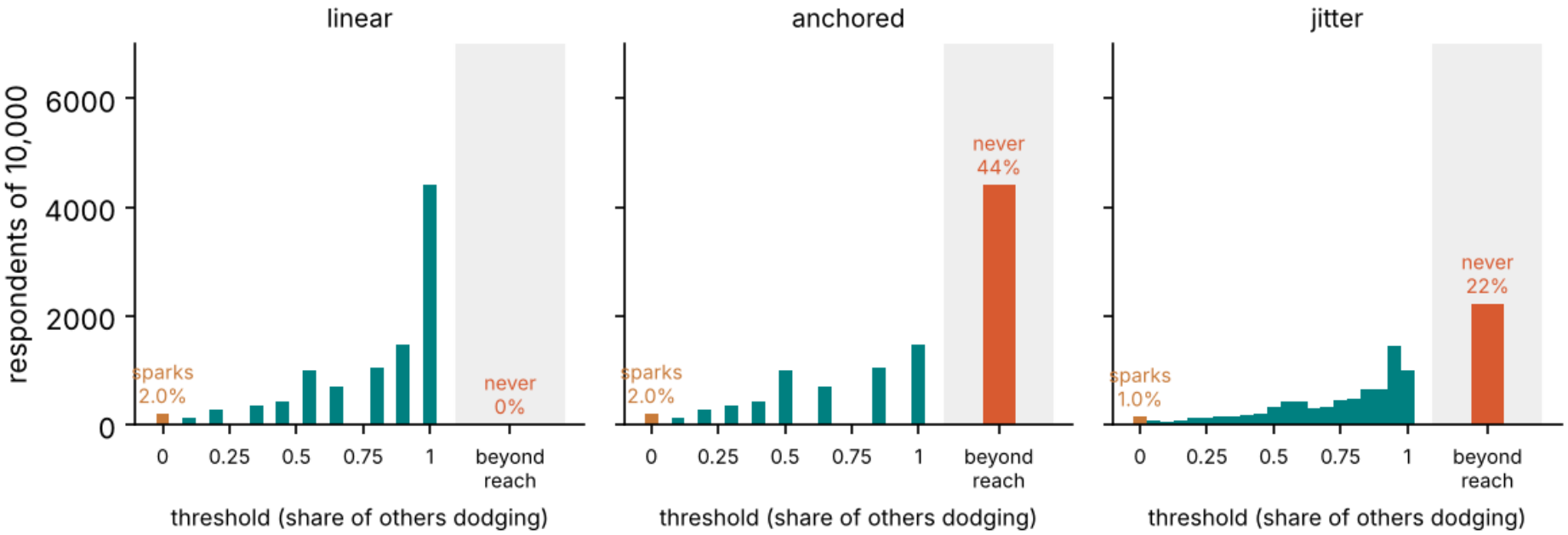
The ends. The 10s' band dips below 0: that part is set to 0, so half are exact sparks (1%). The 1s' band rises above 1: that part is beyond reach, so half the nevers are unreachable (22%).

On a 1–5 scale the step is 1/4, so the band is ± 1/8 around (5 − s)/4: a 3 becomes anything between 0.375 and 0.625. The choice in this rule is the width of the band; one step is the natural width, but it is still a choice.

Four rules compared (the fourth comes in a minute)

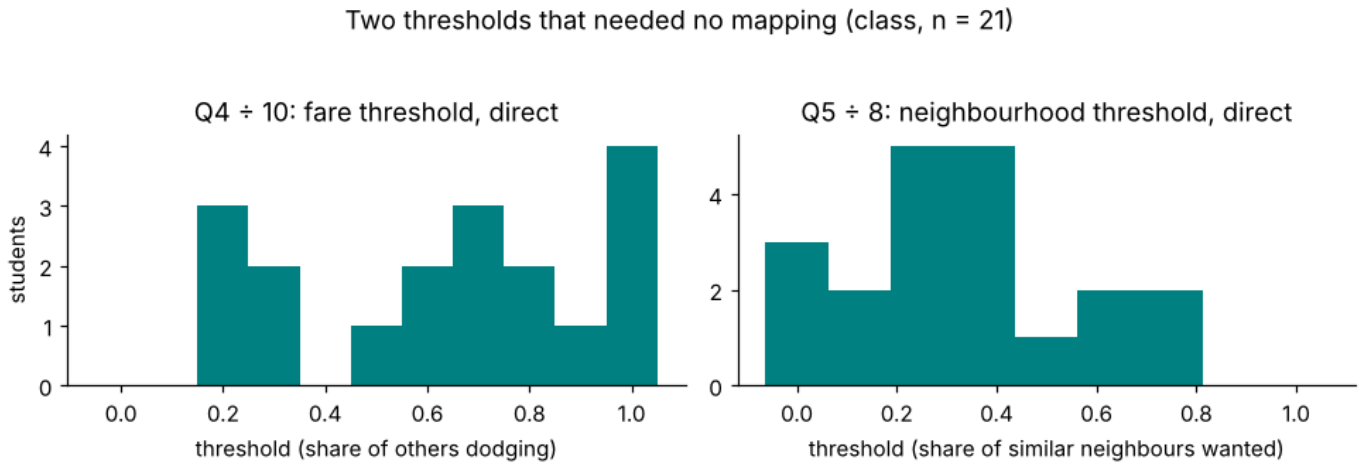
Rule	Assumes	“Never” becomes	Midpoint 5	Strength	Weakness	Use when
Linear	equal distances	1.0, reachable	0.56	simple; nothing hidden	makes “never” reachable	a first pass
Anchored	a few points mean what their words say	beyond 1	0.5, the majority	respects the words	the anchors are a judgement	the end labels and midpoint carry meaning
Jitter	order only; distances unknown	a band, half beyond 1	0.50–0.61	admits what the scale does not say	band width is a choice	the default when nothing better is known
Calibrated	a paired direct question exists	the share the data say	what the data say	learned, not assumed	needs the pair and enough people	whenever you can design the pair in

What the model gets: the same 2011 answers, three rules



10,000 imaginary respondents drawn from the real shares, the same under every rule. Amber: the sparks, threshold exactly 0 (2%, 2%, 1%). Coral, in the grey zone: the nevers, beyond reach (0%, 44%, 22%). The teal middle barely moves. Rules disagree about the ends, and the ends decide a cascade: sparks start it, nevers cap it.

Avoiding the rule, way one: ask in the model's units



- Q4 did not ask how justifiable dodging is; it asked at how many dodgers of ten you would dodge. Divide by 10: the threshold itself. No rule, no choice.
- Q5 did the same for the neighbourhood: how many of eight like you; divide by 8. Stop 5 draws from it.
- The classic case: the Detroit Area Study showed cards of streets with different mixes and asked which you would move into (Farley et al., 1978); Clark (1991) tested Schelling with exactly those distributions.
- Left to decide: the three “never” tickers. The honest answer is the anchored rule's: beyond reach.

Farley, R. et al. (1978). “Chocolate city, vanilla suburbs”. Social Science Research, 7(4), 319–344. · Clark, W. A. V. (1991). Residential preferences and neighborhood racial segregation. Demography, 28(1), 1–19.

Way two: ask twice, and let the pair teach you the rule

Who the rule is for. Not the 21 students: they answered Q4, so they already have a threshold and need no rule. The 1,503 Estonians answered only Q3. The class is a calibration sample: from its pairs we learn what each Q3 answer means in threshold units, then apply that to people who only have Q3. As under every rule, two people with the same answer get the same threshold; only where the number came from differs.

1 · Group
by the Q3 answer: all the 1s, all the 2s, … (groups differ in size; some are empty)

2 · Average
inside each group, the direct thresholds Q4 ÷ 10. That average is the number the answer stands for: the rule

3 · Never
the tick-box has no number: keep it out of the average, count the group's share, and send that share beyond reach

4 · Fill
empty answers get a straight line between neighbours, or the last value carried on. An assumption: say so

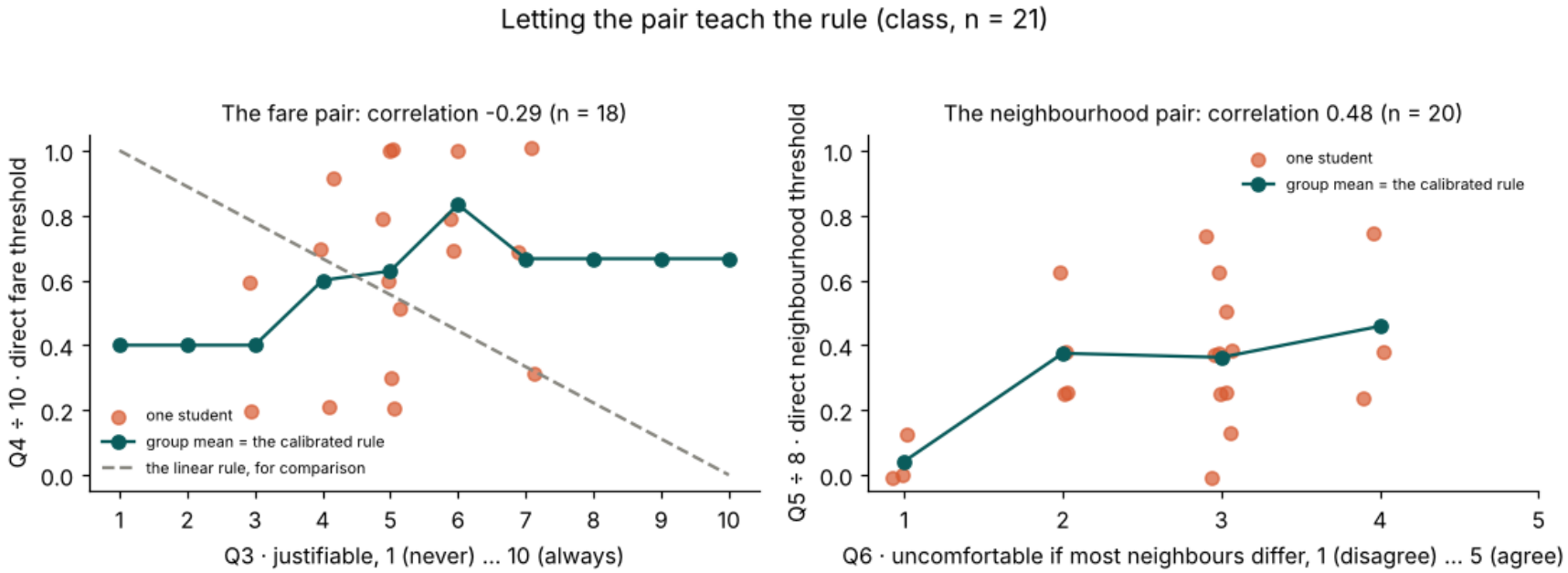
5 · Check
the correlation between Q3 and Q4: does the scale order people the way the direct question does?

6 · Apply
to the 1,503: look up each Estonian's Q3 answer in the table; give them that threshold, or beyond reach with the group's never share

Q3 answer	1	2	3	4	5	6	7	8	9	10
mean Q4 ÷ 10 (the rule)	0.40	0.40	0.40	0.60	0.63	0.83	0.67	0.67*	0.67*	0.67*
share “never” (beyond reach)	1.00	1.00	0	0	0.12	0	0	0*	0*	0*
students the row rests on	1	1	2	3	8	3	3	0	0	0

*The class's own table. * filled: nobody answered 8–10, so the last value is carried on. The students row is the quality control: 0.63 for a 5 rests on eight people; the 0.40 and the never share of 1.00 for a 1 rest on one person, and that one tick-box would send all 659 Estonians who answered 1 beyond reach.*

Way two, applied: the plot, the correlation, the verdict



The plot is steps 1–2 drawn

Each dot a student: Q3 across, Q4 ÷ 10 up. The dark line joins the group averages: the line is the rule. Dashed: linear, for comparison. Where the dark line runs below it, people with that answer are easier to reach than equal spacing assumes.

The correlation is step 5

Pearson correlation between Q3 and Q4 ÷ 10, sign flipped so that "the scale agrees with the threshold" reads positive. Above about 0.6: order right, distances noisy, use the rule. Below about 0.3: the two questions measure different things; do not map.

The verdict for this class

Fare pair: the line rises instead of falling, −0.29: do not map Q3. (Applied anyway, it would send 60% beyond reach on one tick-box and create no sparks.) Neighbourhood pair: 0.04 / 0.38 / 0.36 / 0.46, correlation 0.48: usable, with a wide band.

STOP 2 · CALIBRATION

On your screen: Stop 2

Website tour Stop 2 of the tour: item, year and rule dropdowns; the rule drawn on the left, what the model gets on the right

Notebook Notebook 2.2 (the mapping dropdowns) and 2.4 (the calibration table and plot from your answers)

Parameter	What it means	What to watch
Rule	Which procedure turns the 1–10 answer into a threshold	The amber (sparks) and coral (never) bars move with the rule; the teal middle barely does. Under linear nobody is beyond reach
Item	Which WVS question	For the bribe item the anchored rule puts three quarters of the population beyond reach
Year	1996 or 2011	Sparks and nevers move in opposite directions between the years

Task: find the rule under which the fare item and the tax item give the most similar threshold distributions, and say what that rule assumed to make them similar.

What to take from Stop 2

- Turning an ordinal answer into a model input is a choice among defensible rules; write the choice down.
- The rules disagree most about the extreme answers, which are the ones that decide whether anything happens in the model.
- The way out: ask in the model's units, or ask twice and let the pair teach you the rule. Either way the output is one threshold per agent.

What this stop adds to the argument: *the first real translation, and the discovery that it is a choice.*

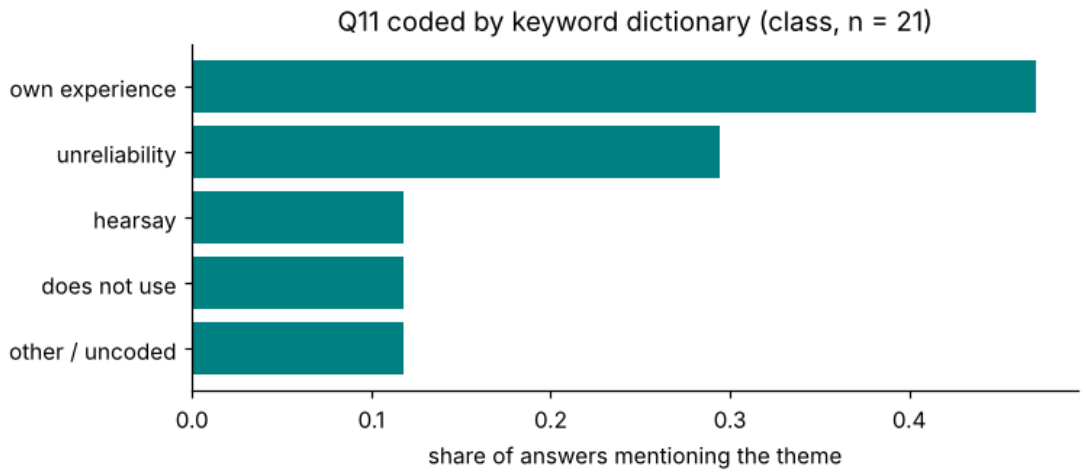
Stop 3

From words and from experts

When there is no scale at all, the translation is still a written rule

Text into categories into shares: your answers to Q11

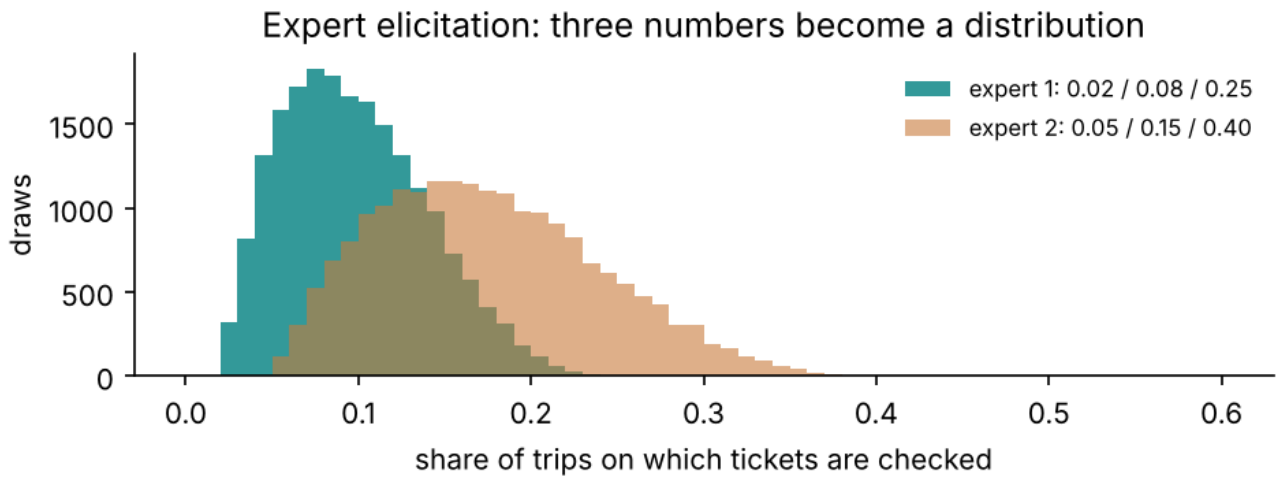
Q11: “In one sentence: why?” (why you trust the timetable as much as you do). 17 sentences. Coding: read each, assign it to one or more themes from a codebook. Counts become shares; shares become probabilities an agent can be drawn with. The notebook uses the crudest coder, a keyword dictionary, so every step is visible.



Theme (keywords)	n	Share
own experience (experience, punctual, stuck to, not failed, trust it …)	8	47%
unreliability (late, early, delay, dont stick, never arrive …)	5	29%
hearsay (heard, feedback)	2	12%
does not use (never used, never travelled …)	2	12%
other / uncoded	2	12%

Three sentences and how they were coded. “From experience, it's punctual” → own experience. “Have missed most of the busses because they come early and leave early” → unreliability. “Because they feel secure answers” → uncoded. Shares sum to more than 100% because a sentence can carry two themes. The uncoded 12% is the first number to report; two independent coders and their agreement would be the second.

A quantity nobody surveyed: expert elicitation



- How often is a ticket checked? No survey gives it. Ask people who know the system for three numbers: lowest plausible, most likely, highest plausible.
- The PERT distribution turns the triple into a bell between lowest and highest, peaked at most likely, leaning to the side with more room; mean (l + 4m + h)/6. A confident expert gives a narrow range.
- Two experts disagreeing is normal; mix their distributions. The overlap is where they agree; the 90% range of the mixture is what a report carries, and the disagreement is the uncertainty to carry forward.

O'Hagan, A. et al. (2006). Uncertain Judgements: Eliciting Experts' Probabilities. Wiley.

On your screen: Stop 3

Website tour Stop 3 of the tour: three sliders for expert 1, a second-expert toggle, and an editable codebook

Notebook Notebook 3.1 (the coder) and 3.2 (the elicitation sliders)

Parameter	What it means	What to watch
Lowest / Most likely / Highest	Expert 1's three numbers	The peak follows “most likely”; narrow the range: a tall thin shape; widen it: a long tail and a wider 90% range
Second expert	A second triple, 5% / 15% / 40%	Move expert 1 so the shapes no longer overlap: the mixture becomes two-humped, and a report should say so rather than average it away
Codebook	Themes and keywords, one per line	The uncoded share. Add a keyword and watch it fall; that is what a codebook is for

Task: elicit the inspection rate from someone who takes the bus daily and from someone who never does, and compare their two shapes.

What to take from Stop 3

- When there is no scale, the translation is still a written rule: a codebook, an elicitation protocol.
- Each produces a distribution, not a number.
- Each has a quality measure (uncoded share, inter-coder agreement, disagreement between experts), and that measure is the uncertainty to carry forward.

What this stop adds to the argument: *the same discipline applied to material that does not even look like data.*

Five-minute break

Next: four kinds of uncertainty, and which one dominates

Stop 4

Quantifying the uncertainty

Sampling, translation, heterogeneity, structural: four kinds, and which one dominates

Four sources, four kinds

Source	What it is	Where it came from today	How it enters the loop
Sampling	you asked n people, not everyone	WVS n ≈ 1,500; class n = 21	resample the respondents (the bootstrap)
Translation	the rule was a choice	linear / anchored / jitter / calibrated; the codebook; the elicitation	run the model under each rule
Heterogeneity	people differ; a population is a distribution, not a point	the spread of thresholds	give each agent its own draw: not error, the phenomenon
Structural	the rule of the model, or the world, could be different	inspections in the rule or not; free fares changing what the item means	compare rule variants; state the condition

Sampling and translation are epistemic: limited knowledge and researcher choices; with unlimited money and a perfect mind they shrink to zero. Heterogeneity is aleatoric: the diversity of the world itself; even a researcher who knew everything would face it. Structural is about the system: it cannot be put in a loop, only stated.

Two definitions, on one statistic

Sampling uncertainty

A different 1,503 Estonians would have given a somewhat different list. Measure it by the bootstrap: draw an imaginary survey of size n from the shares, compute the statistic, repeat 300 times. The spread shrinks like $1/\sqrt{n}$.

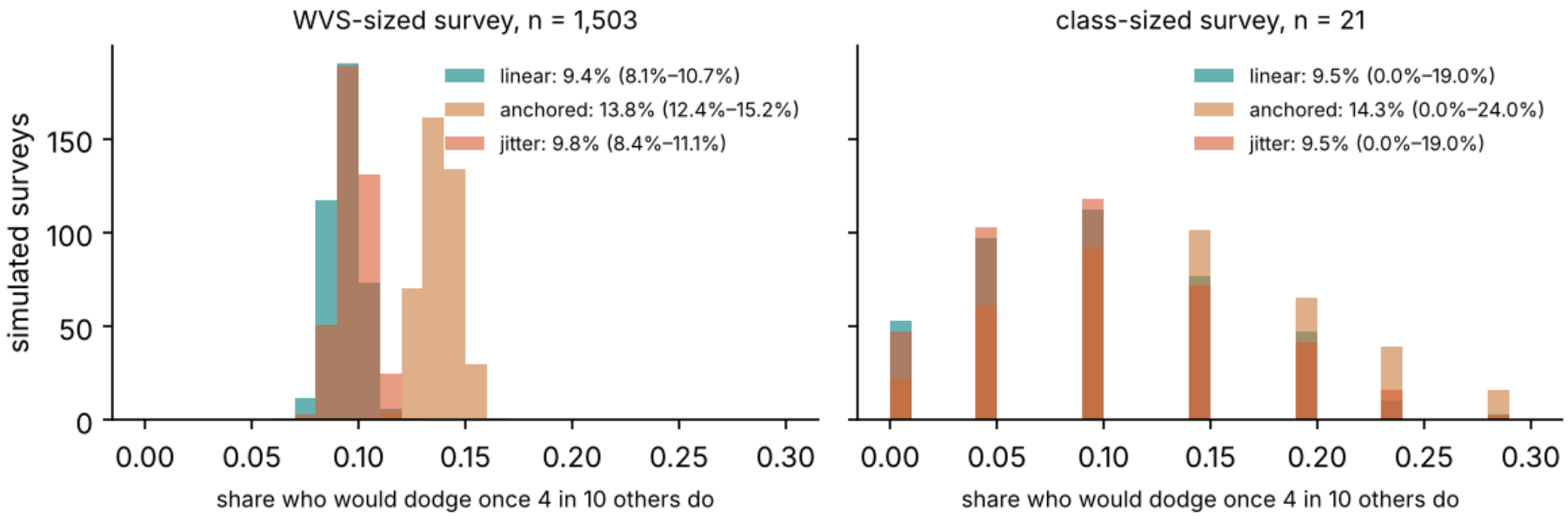
Translation uncertainty

Keep the survey fixed, change only the rule: the statistic moves. The distance it moves does not shrink with n at all, because every extra respondent is translated by the same disputed rule.

The statistic: the share of people who would dodge once 4 in 10 others do (threshold ≤ 0.4). A stand-in for the cascade, fast enough to compute thousands of times. Because thresholds come from a 1–10 scale, it is really a question about scale points:

Rule	Threshold of an answer of 6	Counted at level 0.4?	Answers counted
Linear	0.44	no	7, 8, 9, 10
Anchored	0.40	yes	6, 7, 8, 9, 10
Jitter	0.39 to 0.50	about a tenth of them	7–10 and a few 6s

Sampling against translation, side by side



400 imaginary surveys, each translated by three rules. Width of a colour = sampling uncertainty; distance between colours = translation uncertainty. Left, n = 1,503: narrow histograms in different places: translation dominates, and more people would not move the centres together. Right, n = 21 (imaginary surveys of the class's size drawn from the Estonian shares, not the class's answers): the colours lie on top of one another: sampling dominates, and arguing about the rule is pointless until more people are asked.

Heterogeneity is not noise; structure cannot be averaged

Heterogeneity is not noise

A survey mean summarises a population; a model agent is one person. Give every agent the mean and the model contains clones, and none of the mechanisms that depend on who moves first can operate. The threshold distribution is not uncertainty about the population; it is the population. Stop 5 shows what happens when it is replaced by its mean, in two models.

Structural: what a number cannot carry

Free fares in Tallinn (2013) and the counties (2018) changed what the WVS question means for a large share of Estonians, and no rule fixes that. Say which rule and which world the number belongs to, and report the result as conditional on it. Stop 5 meets a second structural assumption, the passenger without memory, and treats it the same way.

On your screen: Stop 4

Website tour Stop 4 of the tour: survey size, level and year; 300 bootstrap surveys under three rules

Notebook Notebook 4.2: the same, with the bands drawn for n = 1,503 and n = 29

Parameter	What it means	What to watch
Survey size n	How many people each imaginary survey asks	The histograms narrow like 1/√n but their centres do not move: more people never shrink the gap between rules
Others dodging (level)	The share of others at which we count a person as reachable	At some levels the rules coincide, at others one scale point switches sides and they split. The gap is a property of the level, not a constant
Year	1996 or 2011	The rules disagree about the ends, and the ends moved between the years

Task: find the n at which the sampling band becomes narrower than the gap between rules. Beyond it, more interviews stop helping and a better question is the only way forward.

What to take from Stop 4

- Sampling uncertainty shrinks with n; translation uncertainty does not. Measure both on the statistic you need and see which is larger.
- Heterogeneity is the phenomenon, not an error around it. Structural assumptions must be stated, not averaged.
- Find out which uncertainty dominates before spending effort on the other one.

What this stop adds to the argument: *the bookkeeping that makes a translated number honest. A result is reported with the band from every source, and with the sentence that says which world it belongs to.*

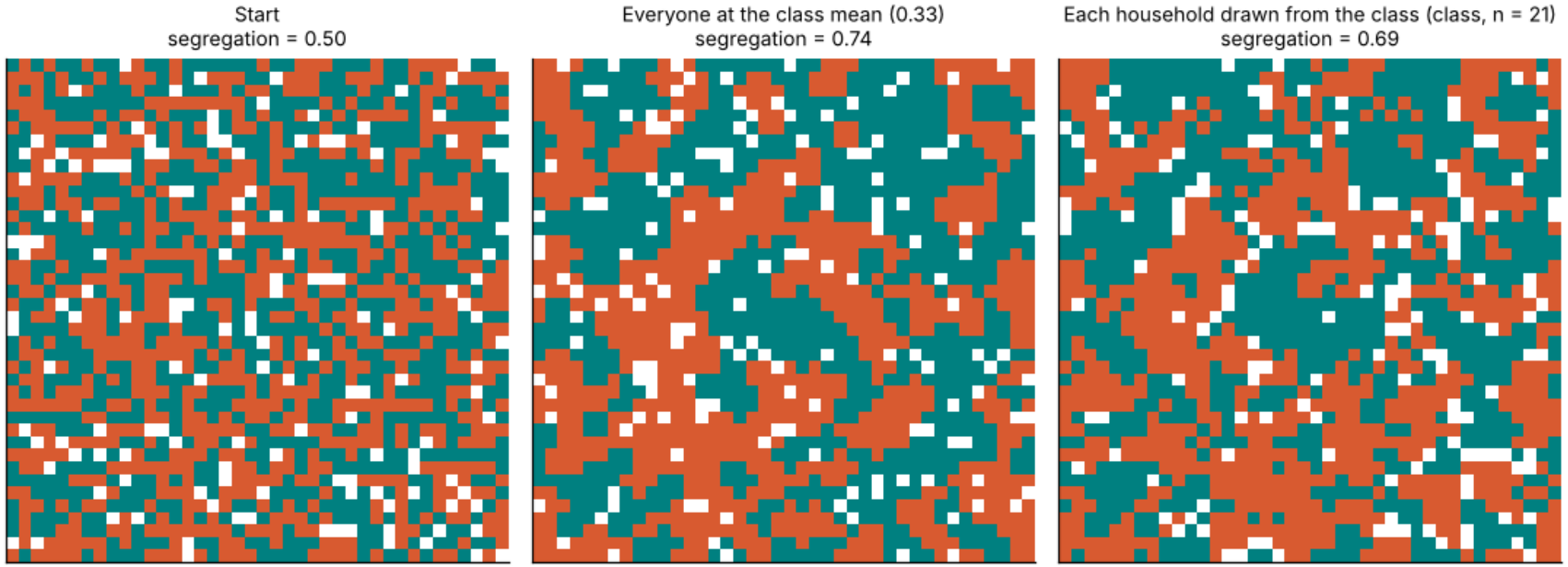
Stop 5

Into the model: distributions in place of numbers

One changed line, and the model acquires behaviour nobody wrote into it

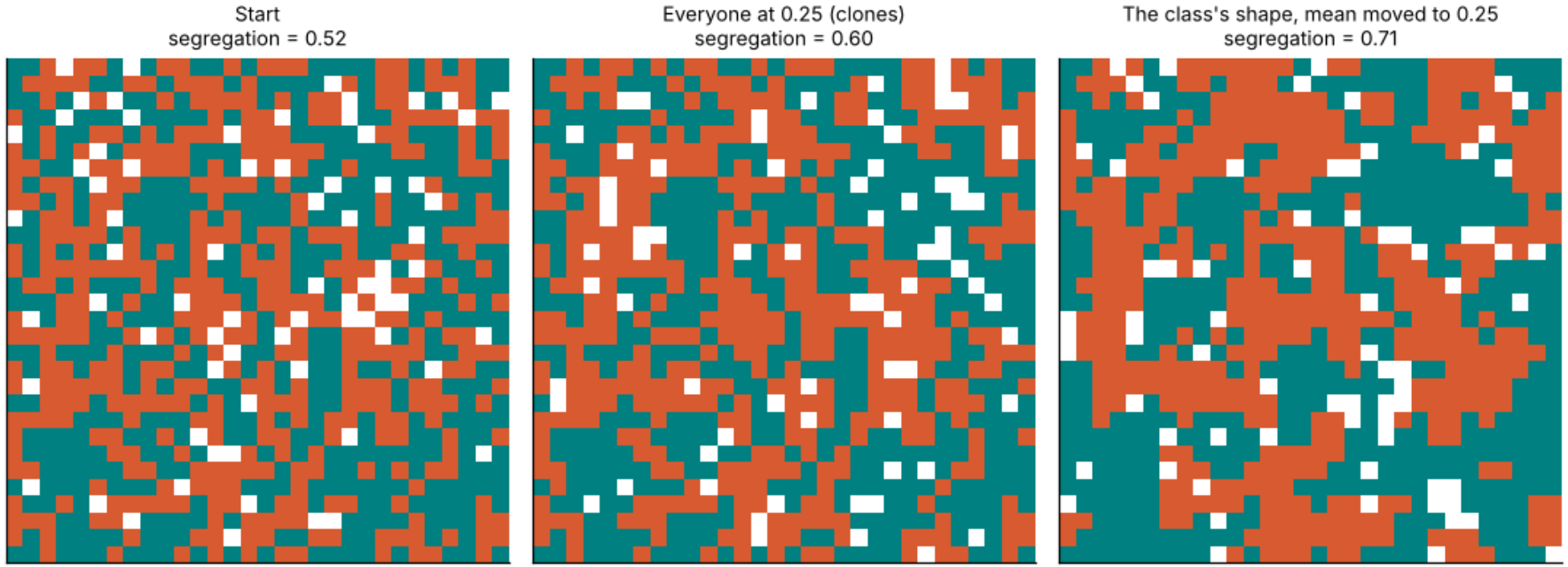
Schelling's city: the one changed line

```
same_for_all   = np.full(city.shape, class_thresholds.mean())    # one number for everyone: the class MEAN
from_the_class = rng.choice(class_thresholds, size=city.shape)    # THE ONE CHANGED LINE: a draw per household
```



Left: the start, 0.50. Middle: every household at the class mean of 0.33: 0.74. Right: each household holds one real student's answer (Q5 ÷ 8), drawn with replacement, so each answer is used by about 75 households: 0.72. The two finished cities look alike, and that is a finding: the class mean sits where even clones segregate.

Where the spread matters: the same mean, below the clones' tipping point



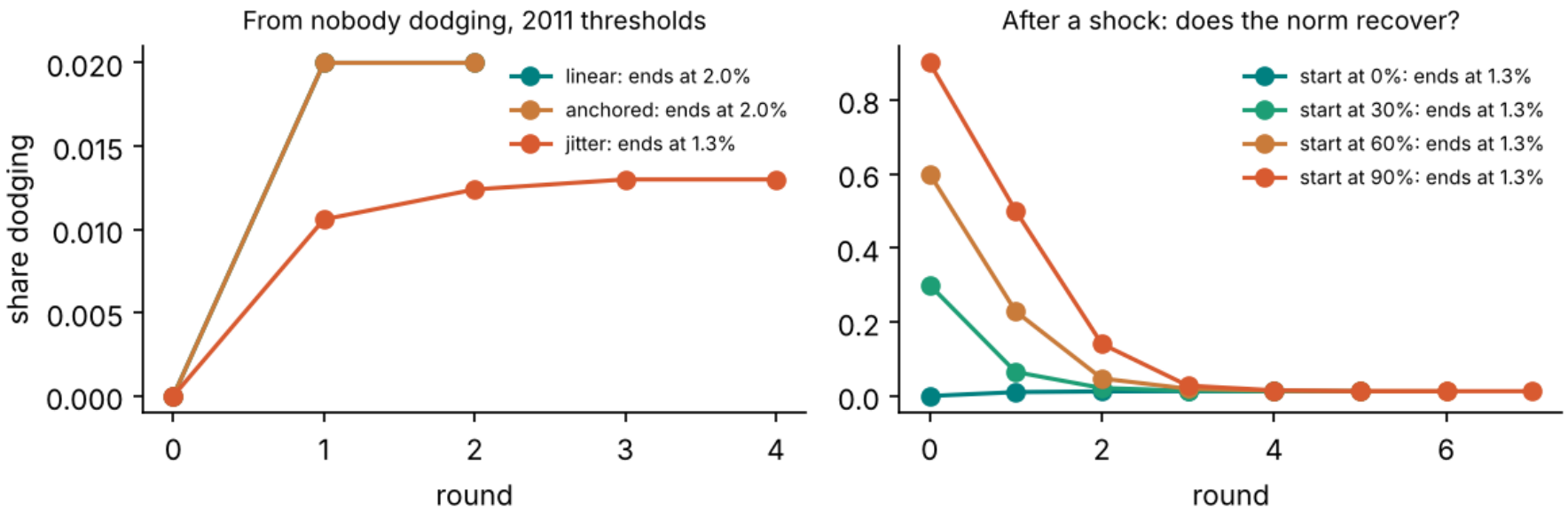
The class's shape moved to a mean of 0.25, below the clones' tipping range. Clones at 0.25 barely segregate; the class's shape around the same mean does, visibly and in the number. A demanding minority keeps moving, clusters form behind it, and the easy-going majority, once inside a cluster, has no reason to leave.

Message: segregation can set in before the average household's threshold says it should. Heterogeneity is not noise around the mean; it is what starts the process.

The fare norm as a cascade: the mechanism

- 5,000 passengers, each with a threshold from Stop 2: the share of others dodging at which this person would dodge too.
- Rounds. Each round every passenger looks at the current share of dodgers and dodges next round if it is at or above their threshold. The new share is the proportion whose threshold the old share reached. Repeat until nothing changes. Granovetter's crowd with Estonian thresholds in it.
- One parameter from Stop 3: the inspection rate. At 12% (the experts' most likely value, the regime the survey was answered under) thresholds are as surveyed; rarer inspections lower every threshold, more frequent ones raise it. The weight (7) has no data behind it: a structural assumption, and a slider.
- A worked round, linear rule, 2011: 2.2% answered “always”, threshold 0, they dodge in round 1. The next lowest thresholds are the 9s at 0.11. A share of 2.2% does not reach 0.11. Nobody else joins; the cascade stops at 2.2%. The sparks are there; the ladder has a missing rung right above them.

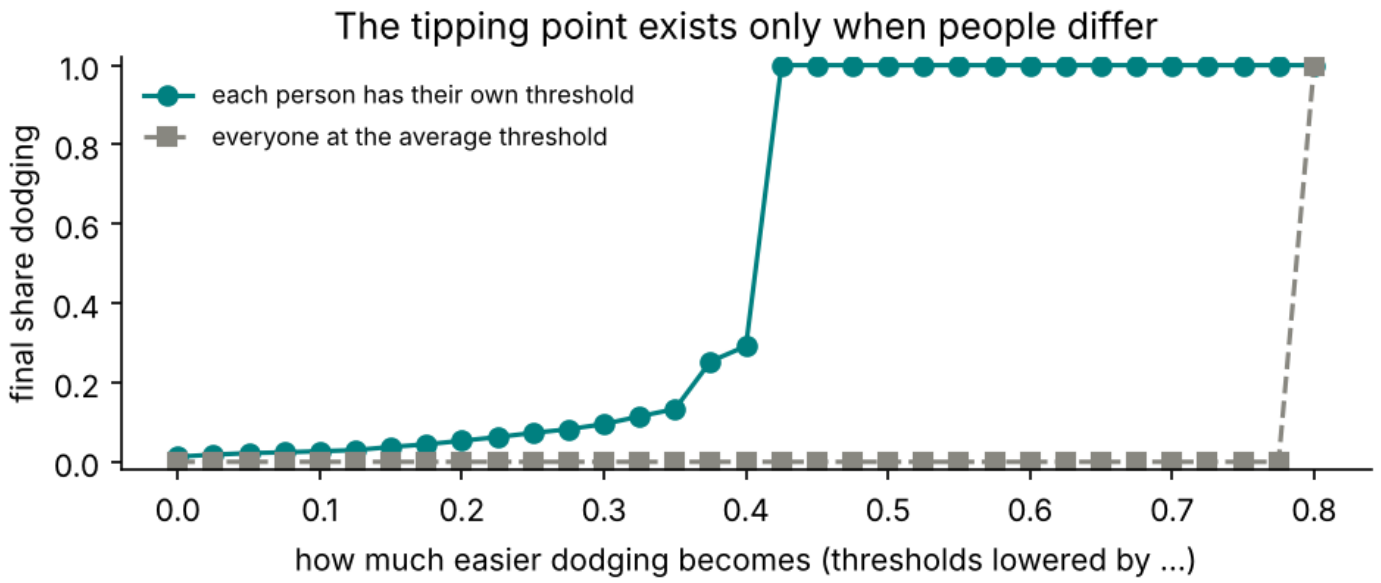
The norm holds, and self-corrects



Left (vertical axis to 2%): from a quiet start, under all three rules, a few per cent and stop. Right (axis to 100%): after a shock of 30%, 60% or 90%, back to the same 1.3% within a few rounds. Conditional on passengers without memory, who decide afresh each round.

Message: the thresholds are exactly Stop 2's distributions. The norm holds under every rule, so at this setting the result does not depend on the translation uncertainty. It self-corrects after any shock, which the strain snapshot of Stop 1 could not show: there each person was judged alone; here they react to each other.

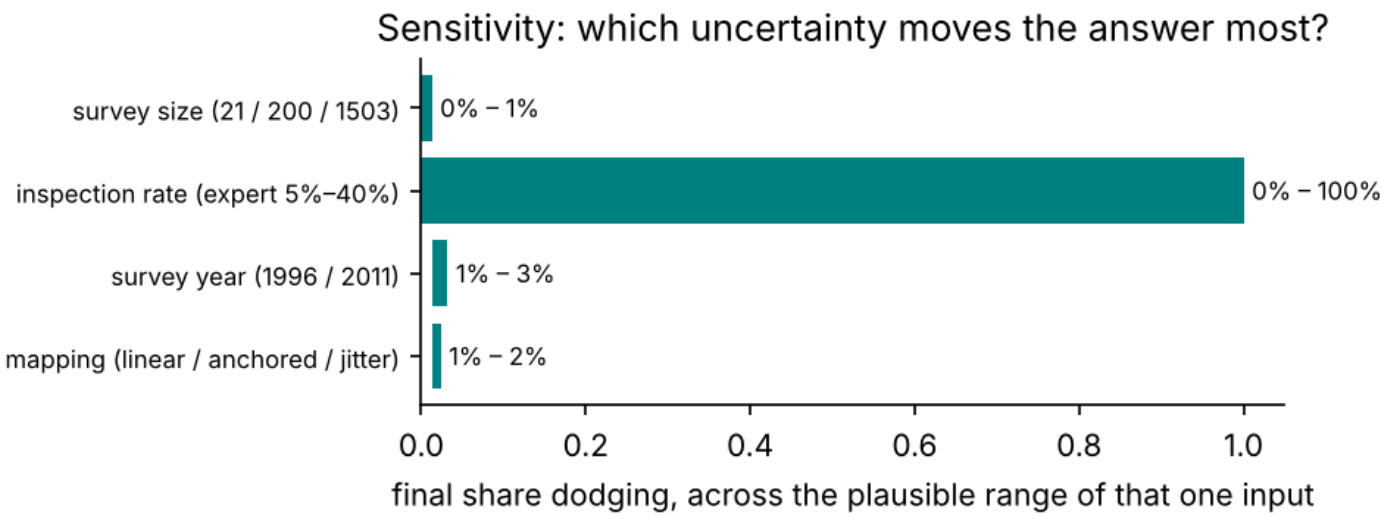
Where it tips, and why the average citizen never tells you



- How each point was made: subtract the shift from every threshold, run the whole cascade from a quiet start, keep its last frame (the final share). 33 times. The previous figure is the film; this is the last frame of 33 films.
- Teal, own thresholds: a visible rise from 2% to 30%, then the rungs become continuous and the crowd tips, near 0.4. The rise is a warning a transport authority could watch.
- Grey, everyone at the average (0.78): nothing until the shift reaches 0.78, then everyone at once. The clones collapse too, in the wrong place and without warning.

Message: the tipping point exists, and sits where it sits, only because people differ. A model that averages them away puts the collapse in the wrong place and removes the warning before it.

Which uncertainty moves the answer? Sensitivity, one input at a time



- Base: 2011, jitter, inspection 12%, crowd of 1,503. Vary one input over its plausible range, others at base, one cascade per value; the bar spans the final shares. Sixteen runs.
- Inspection, 5%–40%: at the low end every threshold drops by 7 × (0.12 − 0.05) = 0.49, past the tipping point. The plausible range of one input straddles the collapse.
- Rule, year, survey size: short bars. At this setting the three rules land on the same side of the tip.
- Move the base near the tipping point and the mapping bar becomes the longest. Sensitivity is a habit, not a step.

What it means for us: at these settings the inspection rate is the uncertainty worth money; a better survey or a better mapping would be wasted effort. That is a truth about this model at this setting, not about mappings in general.

On your screen: Stop 5

Website tour Stop 5 of the tour: the city (mean, spread, seed); the cascade (year, rule, inspection, weight, shock, average citizen; compare buttons); the animated sweep

Notebook Notebook 5.1–5.4: the same cells, plus the full-control cascade

Parameter	What it means	What to watch
Mean / Spread (city)	Average household's demand; how much households differ around it	Spread 0: find the clones' tipping mean. Mean just below it, raise the spread: segregation appears that clones never showed. Notebook: swap in your movie-night or app thresholds
Inspection rate	Share of trips checked; 12% = as surveyed	Lower it step by step and find where the final share jumps: the tipping point for this crowd
Average citizen	Every passenger at the mean threshold	Flat at 0 or a jump to 1, nothing in between; the tipping point you found is gone

Task: find the inspection rate at which the 2011 crowd collapses under jitter; switch to anchored and look again; then turn on the average citizen and look a third time. (The weight slider moves the tipping point: structural uncertainty made visible.)

What to take from Stop 5

- With distributions in place of numbers, the model acquires behaviour nobody wrote into it: self-correction, a tipping point, sensitivity to how people differ.
- Replace the distribution by its mean and the tipping point moves to the wrong place or disappears; the error can go either way.
- Which uncertainty matters depends on where the model sits; check again whenever the setting changes.

What this stop adds to the argument: *the payoff. The translation that looked like bookkeeping is what lets the model produce emergent behaviour; the average citizen never riots, never moves house, and never tells you where the tipping point is.*

Stop 6

Reporting, and your project

Six things a reader needs for every translated number

What to report, for every translated input

Report	For the fare model
The material: what was measured, on what level, from whom, how many	WVS “justifiable” item, ordinal 1–10, Estonia 2011, n = 1,503; class Q4, direct, n = 21
The rule, and the alternatives you did not choose	jitter; linear, anchored and calibrated also run
The distribution the model received, never a single number	a threshold per agent; 22% beyond reach; 1% sparks. Inspection: PERT mixture, median 12%, 90% range 4–30%
Which uncertainty dominates at the setting you report, and of which kind	the inspection rate (epistemic, two experts); rule, year and n barely matter here
The structural assumptions the result is conditional on	passengers without memory; a weight of 7; the 2011 meaning of the item, before free fares
What would change the conclusion	an inspection rate below about 7%; a rule with habit in it; a crowd with fewer “nevers”

The last line is the one readers remember, and the one that tells you what to measure next.

Two more habits that keep you honest

- Last time: garbage in, gospel out; report the spread; use enough runs; fix the seed; a model is not the world.
- 6. Report the rule you did not choose. If a different defensible translation gives a different answer, say so and show both. A result that survives all four rules is a result; one that holds under one rule is a hypothesis about the rule.
- 7. Never collapse a distribution into its mean before the model runs. Give every agent its own draw. The average citizen does not exist, and a model made of average citizens cannot tip, cascade or segregate.

For your project

Before you write a line of model code

For every input ask:
- attribute, rule, parameter or initial state?
- what level of measurement is the source?
- can the question be asked in the model's units, or asked twice?
- which rule am I choosing, and what is the alternative?
- which of the four uncertainties dominates, and is it one that money can shrink?

Then run the model as a Monte Carlo experiment and report the band, not the number, with the sentence that says which world it belongs to.

Now you: ten minutes of free play

Website tour

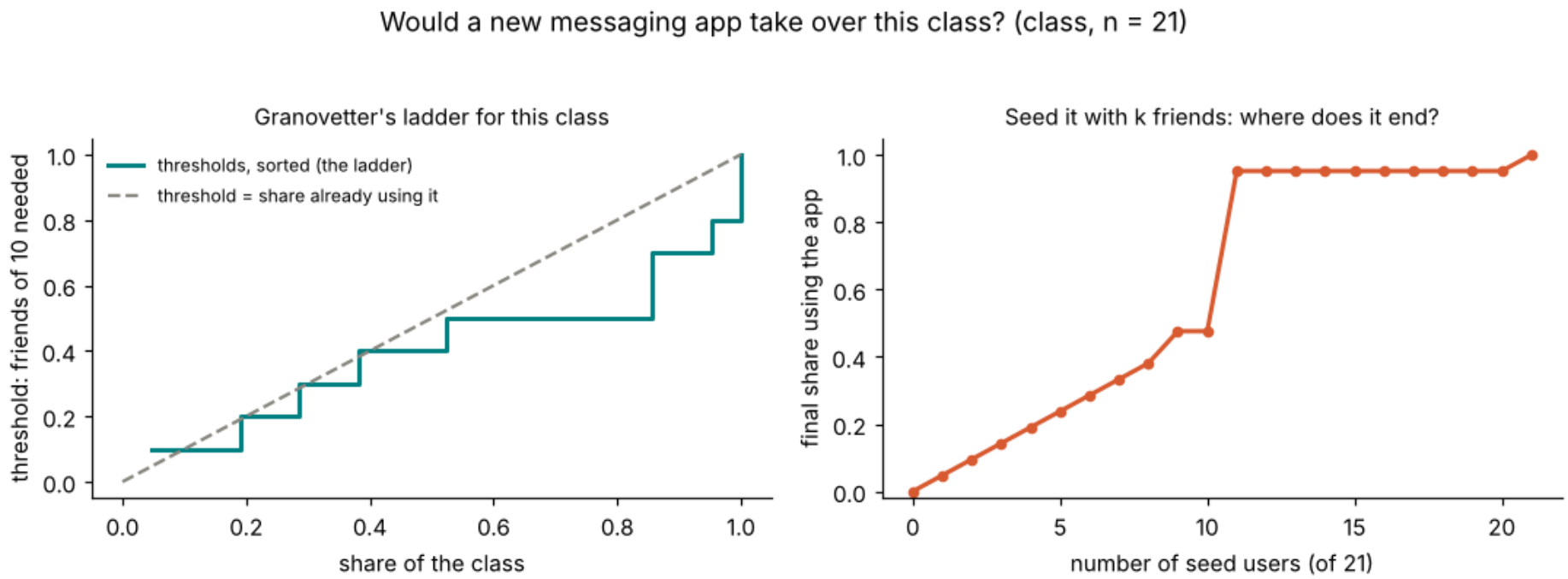
acharyakishor.com/simulations
“From words to numbers”: six stops with sliders, the rule drawn live, the animated sweep, the sensitivity bars with a movable base.

Colab notebook

Open in Colab from github.com/ACHARYAkishor/from_words_to_number_Seminar_UniTartu. Same numbering as these slides; your own answers in the class cells; the real WVS data file with a link per item.

Suggested first tasks: Stop 2, flip the rule and watch only the two coloured bars; Stop 4, find the n where the band becomes narrower than the gap; Stop 5, find the tipping inspection rate, then switch on the average citizen.

For fun: would a new messaging app take over this class?



Q9: at how many of your ten closest friends would you start using a new app? A threshold in the model's units, and the class is a crowd of 21. Seed it with k users (adopters keep it) and run Granovetter's cascade. Left: the thresholds sorted, against the share of the class below each (the ladder); the diagonal is “threshold = share already using it”. Right: the final share for every number of seed users.

Nobody needs zero friends, so nothing starts by itself. Up to half the class, every rung is just out of reach of the one below: 2, 5, 7, 10 seed users all stall. With 11 it runs to 95%, and stops short of the one person who needs all ten friends. No average would have told you that 10 fail and 11 succeed. Notebook 5.5.

References

Bicchieri, C. (2017). Norms in the Wild: How to Diagnose, Measure, and Change Social Norms. Oxford University Press.

Clark, W. A. V. (1991). Residential preferences and neighborhood racial segregation: A test of the Schelling segregation model. Demography, 28(1), 1–19.

Farley, R., Schuman, H., Bianchi, S., Colasanto, D., & Hatchett, S. (1978). “Chocolate city, vanilla suburbs”: Will the trend toward racially separate communities continue? Social Science Research, 7(4), 319–344.

Granovetter, M. (1978). Threshold models of collective behavior. American Journal of Sociology, 83(6), 1420–1443.

Inglehart, R., Haerpfer, C., Moreno, A., Welzel, C., Kizilova, K., Diez-Medrano, J., Lagos, M., Norris, P., Ponarin, E., & Puranen, B. (eds.) (2014). World Values Survey: Round Three (1995–1998) and Round Six (2010–2014), Country-Pooled Datafiles. Madrid: JD Systems Institute. Estonia 1996 (n = 1,021) and 2011 (n = 1,533); frequency tables from catalog.ihsn.org/catalog/9087 and /9049.

Lazarsfeld, P. F. (1958). Evidence and inference in social research. Daedalus, 87(4), 99–130.

O'Hagan, A., Buck, C. E., Daneshkhah, A., Eiser, J. R., Garthwaite, P. H., Jenkinson, D. J., Oakley, J. E., & Rakow, T. (2006). Uncertain Judgements: Eliciting Experts' Probabilities. Wiley.

Schelling, T. C. (1971). Dynamic models of segregation. Journal of Mathematical Sociology, 1(2), 143–186.

Stevens, S. S. (1946). On the theory of scales of measurement. Science, 103(2684), 677–680.

Thank you!

Aitäh!

धन्यवाद

Every number is a translation. Write the rule down. Carry the doubt into the model. Report the band, and the sentence that says which world it belongs to.